

Section 1 - Identification

Product Identifier

Product Name A8 Agar

Recommended Use Laboratory chemicals.
Uses advised against No Information available

Product Code	R20205
Address	Thermo Fisher Scientific New Zealand Ltd 244 Bush Road, Albany, Auckland, New Zealand
Emergency Tel.	CHEMTREC® 09 980 6780 or +64 9 980 6780
Telephone / Fax Numbers	Tel: 09 980 6700 Fax: 09 980 6788
E-mail address	<u>ANZinfo@thermofisher.com</u>

Section 2 - Hazard(s) Identification

Classification under Work Safe New Zealand

Not classified as hazardous according to criteria of EPA New Zealand

GHS Classification

Physical hazards

Based on available data, the classification criteria are not met

Health hazards

Based on available data, the classification criteria are not met

Environmental hazards

Based on available data, the classification criteria are not met

Label Elements None required

Other hazards which do not result in classification

This product does not contain any known or suspected endocrine disruptors

Section 3 - Composition and Information on Ingredients

Component	CAS No	Weight %
Water	7732-18-5	78.16
Agar	9002-18-0	1.25
Calcium chloride, dihydrate	10035-04-8	Trace
HORSE SERUM	RR-36477-2	17.88
Yeast, ext.	8013-01-2	0.22
Adenine (6-Aminopurine)	73-24-5	Trace
6H-Purin-6-one, 2-amino-1,7-dihydro-, monohydrochloride	635-39-2	Trace
Urea	57-13-6	Trace
Penicillin G potassium	113-98-4	Trace
Amphotericin B	1397-89-3	Trace
Vitamin B12	68-19-9	Trace
L-Glutamine	32640-56-5	Trace
p-Aminobenzoic acid	150-13-0	Trace
Cysteine hydrochloride, L-(+)-, monohydrate	7048-04-6	Trace
Glucose	50-99-7	Trace
Manganese sulfate monohydrate	10034-96-5	Trace
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-4-methyl-5-(4,6 ,6-trihydroxy-3,5-dioxo-4,6-diphosphahex-1-yl)-, chloride, P,P'-dioxide	154-87-0	Trace
Iron(III) nitrate nonahydrate	7782-61-8	Trace
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-5-(2-hydroxyet hyl)-4-methyl- chloride, monohydrochloride	67-03-8	Trace
Adenosine 5'-(trihydrogen diphosphate), P'.fwdarw.5'-ester with 3-(aminocarbonyl)-1-.beta.-D-ribofuranosylpyridinium, inner salt	53-84-9	Trace
Hydrogen chloride	7647-01-0	Trace
Tryptic Soy Broth	NA	2.19
1,4-Butanediamine, dihydrochloride	333-93-7	0.16
N2-(N-Glycyl-L-histidyl)-L-lysine monoacetate	72957-37-0	Trace
NONHAZARDOUS	NA	100

Section 4 - First Aid Measures

Description of first aid measures

New Zealand Emergency Tel.	CHEMTREC® 09 980 6780 or +64 9 980 6780
Inhalation	Remove to fresh air.
Eye Contact	Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician.
Skin Contact	Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes.
Ingestion	Clean mouth with water and drink afterwards plenty of water.
Self-Protection of the First Aider	No special precautions required.
First Aid Facilities	Eyewash, safety shower and washroom.
Most important symptoms and effects	No information available.
Notes to Physician	Treat symptomatically.

Section 5 - Fire Fighting Measures

Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Extinguishing media which must not be used for safety reasons

No information available.

Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors.

Hazardous Combustion Products

None under normal use conditions.

Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

Section 6 - Accidental Release Measures

Personal Precautions, Protective Equipment and Emergency Procedures

Emergency procedures

Ensure adequate ventilation.

Environmental Precautions

See Section 12 for additional Ecological Information.

Methods for Containment and Clean Up

Sweep up and shovel into suitable containers for disposal.

Precautions to prevent secondary hazards

Clean contaminated objects and areas thoroughly observing environmental regulations

Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

Section 7 - Handling and Storage

Precautions for Safe Handling

Advice on safe handling

Ensure adequate ventilation.

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

Conditions for Safe Storage, Including any Incompatibilities

Storage Conditions

Keep container tightly closed in a dry and well-ventilated place.

Incompatible Materials

None known.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

Section 8 - Exposure Controls and Personal Protection

Control parameters

Exposure limits

NZ - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

UK - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

ACGIH - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

AUS - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

Component	New Zealand WEL	Australia	ACGIH TLV	The United Kingdom
Vitamin B12				STEL: 0.3 mg/m ³ 15 min TWA: 0.1 mg/m ³ 8 hr Resp. Sens. STEL: 15 mg/m ³ 15 min TWA: 5 mg/m ³ 8 hr Skin
Manganese sulfate monohydrate		TWA: 1 mg/m ³	TWA: 0.02 mg/m ³ TWA: 0.1 mg/m ³	STEL: 0.6 mg/m ³ 15 min STEL: 0.15 mg/m ³ 15 min TWA: 0.2 mg/m ³ 8 hr TWA: 0.05 mg/m ³ 8 hr
Iron(III) nitrate nonahydrate		TWA: 1 mg/m ³	TWA: 1 mg/m ³	STEL: 2 mg/m ³ 15 min TWA: 1 mg/m ³ 8 hr
Hydrogen chloride	Ceiling: 5 ppm Ceiling: 7.5 mg/m ³		Ceiling: 2 ppm	STEL: 5 ppm 15 min STEL: 8 mg/m ³ 15 min TWA: 1 ppm 8 hr TWA: 2 mg/m ³ 8 hr

Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Appropriate engineering controls

Engineering Measures

None under normal use conditions.

Individual protection measures, such as personal protective equipment

Eye Protection

Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

Hand Protection

Protective gloves

Glove material	Breakthrough time	Glove thickness	AUS/NZ Standard	Glove comments
Disposable gloves.	See manufacturers recommendations	-	AS/NZS 2161	(minimum requirement)

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

Skin and body protection

Long sleeved clothing

Respiratory Protection

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use

Recommended Filter type: and maintenance of respiratory protective devices
Particle filter (or AUS/NZ equivalent)

Hygiene Measures Handle in accordance with good industrial hygiene and safety practice.

Environmental exposure controls No information available.

Section 9 - Physical and Chemical Properties

Information on basic physical and chemical properties

Physical State Solid Gel Consistency

Appearance

Odor No information available

Odor Threshold No data available

pH No information available

Melting Point/Range No data available

Softening Point No data available

Boiling Point/Range No information available

Flammability (liquid) No data available

Flammability (solid,gas) No information available

Explosion Limits No data available

Flash Point No information available

Autoignition Temperature No data available

Decomposition Temperature No data available

Viscosity No data available

Water Solubility No information available

Solubility in other solvents No information available

Partition Coefficient (n-octanol/water)

Component **log Pow**

Calcium chloride, dihydrate 0.05

Adenine (6-Aminopurine) -0.1

Urea -1.73

Penicillin G potassium 1.83

Vitamin B12 3.57

Thiazolium, -3.04

3-[(4-amino-2-methyl-5-pyrimidinyl)met

hyl]-5-(2-hydroxyethyl)-4-methyl-

chloride, monohydrochloride

Adenosine 5'-(trihydrogen -4

diphosphate), P'.fwdarw.5'-ester with

3-(aminocarbonyl)-1-.beta.-D-ribofuran

osylpyridinium, inner salt

Vapor Pressure No data available

Density / Specific Gravity No data available

Bulk Density No data available

Vapor Density No data available (Air = 1.0)

Particle characteristics No data available

Other information

VOC Content(%) 0.08

Section 10 - Stability and Reactivity

Reactivity None known, based on information available

Stability	Stable under normal conditions.
Sensitivity to Mechanical Impact	No information available
Sensitivity to Static Discharge	No information available
Hazardous Polymerization	Hazardous polymerization does not occur.
Hazardous Reactions	None under normal processing.
Conditions to Avoid	Heat, flames and sparks.
Incompatible Materials	None known.
Hazardous Decomposition Products	None under normal use conditions.

Section 11 - Toxicological Information

Acute Effects

Information on likely routes of exposure

Product Information	Product does not present an acute toxicity hazard based on known or supplied information
Inhalation	Not an expected route of exposure.
Eyes	Not an expected route of exposure.
Skin	No known effect based on information supplied.
Ingestion	No known effect based on information supplied.

Numerical measures of toxicity

(a) acute toxicity;	
Oral	Based on available data, the classification criteria are not met
Dermal	Based on available data, the classification criteria are not met
Inhalation	Based on available data, the classification criteria are not met

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Water	-	-	-
Agar	LD50 = 11 g/kg (Rat)		
Adenine (6-Aminopurine)	LD50 = 227 mg/kg (Rat)		
Urea	LD50 = 8471 mg/kg (Rat)		
Penicillin G potassium	LD50 = 8900 mg/kg (Rat)		
Amphotericin B	LD50 > 5 g/kg (Rat)		
p-Aminobenzoic acid	>6 g/kg (Rat)		
Glucose	25.8 g/kg (Rat)		
Iron(III) nitrate nonahydrate	LD50 = 3250 mg/kg (Rat)		
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]- 5-(2-hydroxyethyl)-4-methyl- chloride, monohydrochloride	LD50 = 3710 mg/kg (Rat)		
Hydrogen chloride	LD50 238 - 277 mg/kg (Rat)	LD50 > 5010 mg/kg (Rabbit)	LC50 = 1.68 mg/L (Rat) 1 h

(b) skin corrosion/irritation;	No data available
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(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;
 Respiratory No data available
 Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available
 There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available
 Target Organs No information available.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed
 No information available.

Section 12 - Ecological Information

Ecotoxicity

Aquatic ecotoxicity Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

Component	Freshwater Fish	Water Flea	Freshwater Algae	Microtox
Calcium chloride, dihydrate	Lepomis macrochirus: LC50: 10650 mg/L/96h	EC50: 3005 mg/L/48h	-	-
Urea	LC50: 16200 - 18300 mg/L, 96h (Poecilia reticulata)	EC50: = 3910 mg/L, 48h Static (Daphnia magna)		= 23914 mg/L EC50 Photobacterium phosphoreum 5 min
Penicillin G potassium	LC50: > 500 mg/L, 96h static (Leuciscus idus melanotus)	EC50: > 1000 mg/L, 48h (Daphnia magna)		
p-Aminobenzoic acid				= 27.4 mg/L EC50 Photobacterium phosphoreum 30 min 15 °C
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]- 5-(2-hydroxyethyl)-4-methyl- chloride, monohydrochloride	LC50 >100 mg/L/96h	EC50 >100 mg/L/48h		

Terrestrial ecotoxicity There is no data for this product

Persistence and Degradability No information available

Bioaccumulative Potential No information available

Component	log Pow	Bioconcentration factor (BCF)
Calcium chloride, dihydrate	0.05	No data available
Adenine (6-Aminopurine)	-0.1	No data available
Urea	-1.73	<10 dimensionless
Penicillin G potassium	1.83	No data available
Vitamin B12	3.57	No data available
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]- 5-(2-hydroxyethyl)-4-methyl- chloride, monohydrochloride	-3.04	No data available
Adenosine 5'-(trihydrogen diphosphate), P'.fwdarw.5'-ester with 3-(aminocarbonyl)-1-.beta.-D-ribofuranosyl pyridinium, inner salt	-4	No data available

Mobility No information available. .

Other adverse effects

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors
Persistent Organic Pollutant This product does not contain any known or suspected substance
Ozone Depletion Potential This product does not contain any known or suspected substance

Section 13 - Disposal Considerations

Waste treatment methods

Waste from Residues/Unused Products Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

Contaminated Packaging Empty remaining contents. Dispose of in accordance with local regulations. Do not re-use empty containers.

Other Information Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations .

Section 14 - Transport Information

Component	Hazchem Code
Iron(III) nitrate nonahydrate 7782-61-8 (Trace)	2X
Hydrogen chloride 7647-01-0 (Trace)	2RE 2R

NZS 5433:2020 Not regulated

IATA Not regulated

IMDG/IMO Not regulated

Environmental hazards	No hazards identified
Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code	Not applicable, packaged goods
Special Precautions	No special precautions required. Please refer to the applicable dangerous goods regulations for additional information.
Additional information	None known

Section 15 - Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

National Regulations

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

Certified handlers, tracking and controlled substance license requirements

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

International Regulations

Ozone Depletion Potential This product does not contain any known or suspected substance

Persistent Organic Pollutant This product does not contain any known or suspected substance

Rotterdam Convention (PIC) Not applicable

Component	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	IMDG Marine Pollutant
Hydrogen chloride	25 tonne	250 tonne	

Authorisation/Restrictions according to EU REACH

Component	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Calcium chloride, dihydrate	-	Use restricted. See item 75. (see link for restriction details)	-
p-Aminobenzoic acid	-	Use restricted. See item 75. (see link for restriction details)	-
Hydrogen chloride	-	Use restricted. See item 75. (see link for restriction details)	-

<https://echa.europa.eu/substances-restricted-under-reach>

International Inventories

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan

(ISHL), Canada (DSL/NDL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	NZIoC	AICS	EINECS	ELINCS	NLP	KECL	IECSC	TCSI
Water	7732-18-5	X	X	231-791-2	-	-	KE-35400	X	X
Agar	9002-18-0	X	X	232-658-1	-	-	KE-00275	X	X
Calcium chloride, dihydrate	10035-04-8	X	X	-	-	-	-	X	X
HORSE SERUM	RR-36477-2	-	-	-	-	-	-	-	-
Yeast, ext.	8013-01-2	X	X	232-387-9	-	-	KE-05-135 5	X	X
Adenine (6-Aminopurine)	73-24-5	X	X	200-796-1	-	-	KE-29916	X	X
6H-Purin-6-one, 2-amino-1,7-dihydro-, monohydrochloride	635-39-2	X	X	211-235-5	-	-	-	X	X
Urea	57-13-6	X	X	200-315-5	-	-	KE-35144	X	X
Penicillin G potassium	113-98-4	X	-	204-038-0	-	-	KE-11721	-	X
Amphotericin B	1397-89-3	X	X	215-742-2	-	-	-	X	X
Vitamin B12	68-19-9	X	X	200-680-0	-	-	KE-11218	X	X
L-Glutamine	32640-56-5	-	-	-	-	-	-	-	-
p-Aminobenzoic acid	150-13-0	X	X	205-753-0	-	-	KE-01199	X	X
Cysteine hydrochloride, L-(+)-, monohydrate	7048-04-6	X	X	-	-	-	KE-01430	X	X
Glucose	50-99-7	X	X	200-075-1	-	-	KE-17727	X	X
Manganese sulfate monohydrate	10034-96-5	X	X	-	-	-	-	X	X
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-4-methyl-5-(4,6,6-trihydroxy-3,5-dioxo-4,6-diphosphahex-1-yl)-, chloride, P,P'-dioxide	154-87-0	X	X	205-836-1	-	-	KE-01484	X	X
Iron(III) nitrate nonahydrate	7782-61-8	X	X	-	-	-	-	X	X
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-5-(2-hydroxyethyl)-4-methyl- yl- chloride, monohydrochloride	67-03-8	X	X	200-641-8	-	-	KE-01482	X	X
Adenosine 5'-(trihydrogen diphosphate), P'.fwdarw.5'-ester with 3-(aminocarbonyl)-1-.beta.-D-ribof uranosylpyridinium, inner salt	53-84-9	X	X	200-184-4	-	-	KE-25879	X	X
Hydrogen chloride	7647-01-0	X	X	231-595-7	-	-	KE-20189	X	X
Tryptic Soy Broth	NA	-	-	-	-	-	-	-	-
1,4-Butanediamine, dihydrochloride	333-93-7	X	-	206-375-9	-	-	-	-	X
N2-(N-Glycyl-L-histidyl)-L-lysine monoacetate	72957-37-0	-	-	277-125-4	-	-	-	-	X
NONHAZARDOUS	NA	-	-	-	-	-	-	-	-

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDL	PICCS	ISHL	ENCS
Water	7732-18-5	X	ACTIVE	X	-	X	-	X
Agar	9002-18-0	X	ACTIVE	X	-	X	-	-
Calcium chloride, dihydrate	10035-04-8	-	-	-	-	X	X	X
HORSE SERUM	RR-36477-2	-	-	-	-	-	-	-
Yeast, ext.	8013-01-2	X	ACTIVE	X	-	X	-	-
Adenine (6-Aminopurine)	73-24-5	X	ACTIVE	X	-	X	X	X
6H-Purin-6-one, 2-amino-1,7-dihydro-, monohydrochloride	635-39-2	X	ACTIVE	-	X	-	-	-
Urea	57-13-6	X	ACTIVE	X	-	X	X	X
Penicillin G potassium	113-98-4	X	ACTIVE	X	-	X	-	X
Amphotericin B	1397-89-3	-	-	-	-	X	-	X
Vitamin B12	68-19-9	X	ACTIVE	X	-	X	X	-
L-Glutamine	32640-56-5	-	-	-	-	-	-	-
p-Aminobenzoic acid	150-13-0	X	ACTIVE	X	-	X	X	X
Cysteine hydrochloride, L-(+)-, monohydrate	7048-04-6	-	-	-	-	X	-	X
Glucose	50-99-7	X	ACTIVE	X	-	X	X	X
Manganese sulfate monohydrate	10034-96-5	-	-	-	-	X	X	X

Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-4-methyl-5-(4,6-trihydroxy-3,5-dioxo-4,6-diphosphahex-1-yl)-, chloride, P,P'-dioxide	154-87-0	X	ACTIVE	-	X	-	-	-
Iron(III) nitrate nonahydrate	7782-61-8	-	-	-	-	X	-	X
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-5-(2-hydroxyethyl)-4-methyl- chloride, monohydrochloride	67-03-8	X	ACTIVE	X	-	X	X	X
Adenosine 5'-(trihydrogen diphosphate), P'.fwdarw.5'-ester with 3-(aminocarbonyl)-1-.beta.-D-ribof uranosylpyridinium, inner salt	53-84-9	X	ACTIVE	X	-	-	-	-
Hydrogen chloride	7647-01-0	X	ACTIVE	X	-	X	X	X
Tryptic Soy Broth	NA	-	-	-	-	-	-	-
1,4-Butanediamine, dihydrochloride	333-93-7	X	ACTIVE	-	X	X	-	-
N2-(N-Glycyl-L-histidyl)-L-lysine monoacetate	72957-37-0	-	-	-	-	-	-	-
NONHAZARDOUS	NA	-	-	-	-	-	-	-

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

Section 16 - Other Information

This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations

Legend

NZIoC - New Zealand Inventory of Chemicals

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

IECSC - Chinese Inventory of Existing Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

NZS 5433:2020 - Transport of Dangerous Goods on Land

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

WEL - Workplace Exposure Limit

DNEL - Derived No Effect Level

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

VOC - (Volatile Organic Compound)

AICS - Australian Inventory of Chemical Substances

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

CAS - Chemical Abstracts Service

ACGIH - American Conference of Governmental Industrial Hygienists

PNEC - Predicted No Effect Concentration

OECD - Organisation for Economic Co-operation and Development

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

ADG - Australian Code for the Transport of Dangerous Goods by Road and Rail

LC50 - Lethal Concentration 50%

ATE - Acute Toxicity Estimate

RPE - Respiratory Protective Equipment

NOEC - No Observed Effect Concentration

BCF - Bioconcentration factor

PBT - Persistent, Bioaccumulative, Toxic

Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

EPA Guide to classifying hazardous substances in New Zealand

EPA - Assigning a product to an existing HSNO approval guide

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards On basis of test data

Health Hazards Calculation method

Environmental hazards Calculation method

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Revision Date

30-Jun-2023

Revision Summary

Update to CLP Format

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet